

SAKSHAM AGARWAL

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SUMMARY

Architect of production-grade systems spanning a multi-modal deepfake forensics platform with explainable AI, an LLM-powered natural-language app compiler, a real-time collaborative canvas engine, and a hybrid malware detection engine. B.Tech CSE student at VIT Bhopal (CGPA: 8.46/10, graduating 2027) with hands-on expertise in full-stack web development, applied ML/AI systems, and cybersecurity. Proficient across the Python and JavaScript ecosystems; experienced with FastAPI, React/Next.js, and Google Cloud Platform.

EDUCATION

VIT Bhopal University — B.Tech, Computer Science Engineering 2023 – 2027

CGPA: 8.46 / 10.0

St. Anthony's Senior Secondary School — CBSE (Science Stream) 2019 – 2022

TECHNICAL SKILLS

- Programming Languages: Python, Java, C++, JavaScript, TypeScript, C
- Web Development: React.js, Next.js, FastAPI, Flask, Node.js, HTML, CSS, Bootstrap, RESTful APIs
- Machine Learning & AI: PyTorch, scikit-learn, TensorFlow, Computer Vision (OpenCV, MediaPipe), Explainable AI (Grad-CAM, SHAP), Generative AI (LLM Pipelines/API), OCR (Tesseract)
- Database & Cloud: SQL, Firebase, Appwrite, Google Cloud Platform, Cloud Architecture
- Tools & Technologies: Git, GitHub, Docker, CI/CD, YARA, Jupyter Notebook, Linux

FEATURED PROJECTS

Deepfake Forensics & Explainable AI Platform | *Python, PyTorch, FastAPI, React, OpenCV, MediaPipe, EfficientNet-B4* [Code →](#)

- Architected a 15-signal ensemble system for synthetic media detection — EfficientNet-B4 classification, Grad-CAM/SHAP explainability, SyncNet lip-sync analysis, and rPPG/frequency-domain forensics.
- Automated FastAPI + React/Vite report generation into 8-chapter, court-ready PDF dossiers embedding 30+ visual forensic exhibits in a 2-column auto-paginated gallery.
- Achieved 99.81% accuracy (ROC-AUC 1.0000, 100% recall) on the official Celeb-DF v2 benchmark and 99.96% accuracy on 140k Real & Fake Faces, training the visual backbone on 73,373 curated samples.

NL App Compiler | *Node.js, JavaScript, Gemini API* [Code →](#)

- Designed a 4-stage, compiler-style LLM pipeline covering intent parsing, UI/UX generation, schema synthesis, and cross-layer refinement.
- Orchestrated chained Gemini API calls to turn natural-language prompts into validated, deployable web applications end-to-end.

NexusBoard | *React 18, Tailwind CSS, Socket.IO, Node.js, Canvas API* [Code →](#)

- Built a live collaborative infinite canvas over Socket.IO, stress-tested to 50 concurrent users/room (100+ across a cluster) with sub-5ms sync latency (median 0.56ms) and zero packet loss.
- Optimized a 60 FPS offscreen double-buffered Canvas 2D rendering pipeline handling 1,000 strokes (100,000 vertices) per frame in 0.92ms, sustaining 15,718 strokes/sec peak ingestion throughput.

DocPilot – AI-Powered Clinical Management Platform | *React, TypeScript, Firebase, Appwrite, Gemini AI* [Code →](#)

- Developed an end-to-end, role-based clinical platform for doctors and patients with secure Firebase multi-role authentication.
- Integrated Gemini AI as a consultation scribe that auto-generates structured clinical notes and visit summaries, synced instantly via Appwrite.

Malware Detector | *Python, Flask, YARA, VirusTotal API* [Code →](#)

- Engineered a 5-layer static malware triage engine combining 3,791,136 cryptographic hash signatures, 1,586 YARA rule files (20,523+ definitions), and PE Shannon-entropy heuristics with VirusTotal v3 REST telemetry.
- Attained 100% detection on test malware corpora with sub-1ms hash lookups and under 0.5% YARA false-positive rate, secured via Fernet 256-bit encryption and OWASP-hardened HTTP headers.

HACKATHONS & COMPETITIONS

- TCS CodeVita — Round 1 Qualifier — Ranked in the top 33% globally (Rank 6,735 of 20,540) of the international coding contest.
- Gridlock Hackathon 2.0 (Flipkart) — Round 2 Qualifier — Qualified solo in Flipkart's national ML hackathon by building a traffic-demand prediction model achieving 93.94% accuracy on HackerEarth.

PROFESSIONAL CERTIFICATIONS

Generative AI Professional Certificate (IBM, 2025) • AI & Edge Computing for Industry Applications (L&T EduTech, 2025) • Bits and Bytes of Computer Networking (Google, 2025) • [Google Cloud Skills Boost \(2025\)](#) • [AWS Certified Cloud Practitioner](#) • [AWS Certified AI Practitioner](#)

Verify remaining credentials: [LinkedIn – Licenses & Certifications](#)

VOLUNTEERING & LEADERSHIP

Design Team Lead, Fintech Club — VIT Bhopal 2025 – Present

- Spearheaded the club's end-to-end visual identity, designing social media assets and event collateral for all club activities.
- Directed digital campaign design that drove consistent brand presence across the club's online and offline presence.